

A Letter to Peter Gärdenfors: On the Third Level of Representation, Found Independently — and Whether Our Convergence Is Real

Yiwei Zhang (math4mad)
The Cora Project — github.com/math4mad

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Dear Professor Gärdenfors,

I am writing from a small research garden in China, under the hand of a former statistics teacher. I should tell you the whole story, including the part where our lineage turns out to be *independent* of your book — which is why I dare to write to you at all.

Our framework did not descend from your work. It descended, in 2017, from a neuroscience result: Chang & Tsao’s face patches (*Cell*), where ~ 200 neurons encode any face through **50 linear tuning axes** — a face as a coefficient vector over inherited basis functions. My intuition since then has been one sentence: *cognition works like JPEG — built-in bases, memory stores only coefficients*. Your book was on my shelf, unread to the end; perhaps the opening pages seeped in. Only this month, while cross-checking our papers against it with an AI reading companion, did we discover how exact the overlap is — and there is a small comedy in the discovery: asked to find the sentence “*Concept learning is closely tied to the notion of similarity, which has turned out to be problematic for the symbolic and associationistic approaches*,” the machine searched the wrong 656-page PDF (a machine-learning textbook) and pronounced it **not in your book**. It is in your book. Page 10, word for word. We verified every anchor against a scanned copy, line by line:

- p.10** — the complaint above, and your answer: “*I advocate a third form of representing information that is based on using geometrical structures.*”
- p.52** — the three levels: symbolic / conceptual / subconceptual.
- p.136** — “*similarity and distances in conceptual spaces are intimately connected.*”
- p.166** — generalized Voronoi tessellation with prototypical *areas*.
- p.278** — Kohonen networks representing information “*very much like it would be represented in a conceptual space.*”
- p.293** — the three levels as different scales of resolution: high-dimensional vectors at the sub-conceptual level, reduced to low-dimensional structured vectors at the conceptual level.

I tell you this because in 2026 our garden, arriving from the opposite shore (face cells, JPEG, subspace algebra), built the missing middle level you called for — **with measuring instruments**. That two independent derivations land on the same geometry is, we hope, the kind of multiple evidence Popper would have trusted: the conceptual level may be an inevitable property of intelligent systems, not a school’s assumption. Our framework (the Atlas framework;

two papers attached) treats a fine-tuned LoRA adapter's ΔW as a *low-dimensional structured vector* in exactly your p.293 sense:

1. **Concepts as regions** → **concepts as centered low-rank subspaces**. A concept-space is operationally a subspace of weight space; similarity between concept-spaces is the principal angle between them (after global centering, intra/inter separation improves 13-fold — your “similarity as distance”, made measurable on actual neural networks).
2. **Categorization as Voronoi** → **probe collapse**. A token is not an information carrier but a *probe*: it collapses a Dirichlet posterior over spaces, and the argmax region is the categorization. Your generalized Voronoi, run against living language models (0.5B–7B): the geometry of concept-space connectivity improves monotonically with scale.
3. **The stratification** of subconceptual / conceptual / symbolic reappears as weights / adapter-subspace / token-behavior — and we can now *walk between the floors*: read a concept off the weights (SVD of ΔW) and verify it behaviorally (posterior alignment), two instruments on one quantity.
4. **Even your SOM hint** (p.278) has a 2026 successor: adapters trained on narrative vs. permuted vs. cross-domain text differ by a *density law of ordinal structure* (7° / 56° / 83° subspace angles).

We also found, in the same forensic work, a poetic confirmation of your similarity thesis at the level of *biography*: commit-interval rhythms of human-written repositories versus AI-agent-written repositories differ by a factor of 60 (median 1553 minutes vs. 21) — “names can be borrowed, heartbeats cannot.” A geometry of thought, it turns out, has a pulse. The protocol behind such verdicts we call, after the man himself, the **Gram-Schmidt duel**: two agents are compared dimension by dimension, each round projecting out every dimension already contested; the first orthogonal residual decides, and the last dimension carries no position effect.

Three humble asks, with no obligation whatsoever:

1. **Judge the convergence**. Reading our two papers against your 2000 (and 2014) geometry: where the convergence is real, is it *for the right reasons*? Where it is only a homomorphism pretending to be an isomorphism, say so plainly — our ledger will record that too. We falsify our own claims before others do it for us.
2. Did you have a name in mind for the third level's *metrics*? We chose subspace angles; if your later work (including *Geometry of Meaning*, 2014) has preferences or warnings we should heed, we would be glad to cite them.
3. If our reading mistakes your program — the claim that concepts are regions was ours before we knew it was yours — correcting the record publicly is itself a service we would cite with gratitude.

Twenty-six years ago you drew the geometry of a third level of representation and left it without instruments; a decade ago face cells got theirs. That our small independent route — one human, two LLM colleagues, a lot of GPUs' worth of patience — should arrive at the same map is the finding of this letter. If the convergence is real, your book will be cited in ours the way a second sight is cited: not as the source of the walk, but as proof the road was there.

With deep respect,

Yiwei Zhang ([math4mad](https://math4mad.github.io))

The Cora Project — github.com/math4mad

<https://math4mad.github.io/cora-atlas/>

Prose developed with LLM conversational partners (Qwen2.5 + an agent harness); all claims, numbers and errors are the human author's.